MAKING THE COMPONENTS OF THE GENERIC FIBRE GEOMETRICALLY IRREDUCIBLE BY A FINITE ETALE BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 6771-6793 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 6795-6829, the complete author proof (33 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Finite etale coverings, irreducible components of a generic fibre, and geometric irreducibility of a component; the two-step construction is retained.

Source reference. Original lemma 00U6: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6829–6829. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 030N: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6829–6829. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 054R: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6829–6829. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes. Assume that Y is irreducible and f is of finite type. There exists a diagram*

$$\begin{array}{ccccc} X' & \xrightarrow{g'} & X_V & \longrightarrow & X \\ {\scriptstyle f'}\downarrow & & \downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & V & \longrightarrow & Y \end{array}$$

where

(1) *V is a nonempty open of Y,*
(2) *$X_V = V \times_Y X$,*
(3) *$g : Y' \to V$ is surjective finite étale,*

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

(4) $X' = Y' \times_Y X = Y' \times_V X_V$,
(5) g' *is surjective finite étale,*
(6) Y' *is an irreducible affine scheme, and*
(7) *all irreducible components of the generic fibre of* f' *are geometrically irreducible.*

Proof. Let $V = \mathrm{Spec}(A)$ be a nonempty affine open of Y. By assumption the nilradical $\mathfrak{p} = \sqrt{(0)}$ of A is a prime ideal. Let $K = \kappa(\mathfrak{p})$. By Varieties, Lemma 054R there exists a finite separable field extension K'/K such that all irreducible components of $X_{K'}$ are geometrically irreducible over K'. Choose an element $\alpha \in K'$ which generates K' over K, see Fields, Lemma 030N. Let $P(T) \in K[T]$ be the minimal polynomial for α over K. After replacing α by $f\alpha$ for some $f \in A$, $f \notin \mathfrak{p}$ we may assume that there exists a monic polynomial $T^d + a_1 T^{d-1} + \ldots + a_d \in A[T]$ which maps to $P(T) \in K[T]$ under the map $A[T] \to K[T]$. Set $A' = A[T]/(P)$. Then $A \to A'$ is a finite free ring map such that there exists a unique prime $\mathfrak{q}$ lying over $\mathfrak{p}$, such that $K = \kappa(\mathfrak{p}) \subset \kappa(\mathfrak{q}) = K'$ is finite separable, and such that $\mathfrak{p}A'_\mathfrak{q}$ is the maximal ideal of $A'_\mathfrak{q}$. Hence $g : Y' = \mathrm{Spec}(A') \to V = \mathrm{Spec}(A)$ is étale at $\mathfrak{q}$, see Algebra, Lemma 00U6. This means that there exists an open $W \subset \mathrm{Spec}(A')$ such that $g|_W : W \to \mathrm{Spec}(A)$ is étale. Since g is finite and since $\mathfrak{q}$ is the only point lying over $\mathfrak{p}$ we see that $Z = g(Y' \setminus W)$ is a closed subset of V not containing $\mathfrak{p}$. Hence after replacing V by a principal affine open of V which does not meet Z we obtain that g is finite étale. □